Joseph Barbosa

josephbarbosa.com | 647-328-9032 | josephbarbosa416@gmail.com | linkedin.com/in/joseph-c-barbosa | github.com/joeyb007

Education

University of Waterloo & Wilfrid Laurier University GPA: 3.98 / 4.0 (92%)
BCS Computer Science (UW) & BBA Business Administration (WLU) Double Degree *Sep. 2025 – Present*

Experience

Software Engineering Intern May 2026 – Aug. 2026
Scotiabank *Toronto, ON*

- Full-stack development (React, Python, Docker); leading an **agentic QA automation** initiative (LangGraph, Azure Foundry) where an agent analyzes Jira Stories to auto-generate **Playwright** tests. Projected to cut manual QA from **∼2h to ∼30m**
- Engineered the agent to run entirely within the bank's locked-down network (no external cloud APIs or package registries) by self-hosting an LLM via **Foundry Local**, satisfying enterprise security and compliance requirements

Software Engineering Intern Jan. 2026 – Apr. 2026
ESGTree *Waterloo, ON*

- Built a production NL-to-SQL agent (ReAct loop, FAISS schema retrieval, FK-aware join hints) over a **98-table** MySQL DB; Achieved **90.9%** accuracy on a **77**-query eval suite, with ablations crediting self-correction (**+13.0pp**) and RAG (**+7.8pp**)
- Engineered backend with read-only DB privileges, regex SQL validation, and prompt-level constraints; enforced **multi-tenant isolation** via JWT; architected a **3-model cost/latency tier** (Sonnet reasoning, Opus codegen, Haiku summarization)
- Extended the agent with Docker-sandboxed Python execution (pandas/scipy/sklearn) for trend projection and statistics

Software Development Intern Jun. 2024 – Dec. 2024
FuturIQ *Brampton, ON*

- Consulted with realtors to pilot, build, and deploy an end-to-end production ML pipeline for housing price prediction: **Selenium** ingestion of **1,000+** listings, **8** engineered features, and a cross-validated regularized polynomial model serving **real-time predictions to clients** via a Flask/React app on Heroku with automated retraining on freshly scraped market data

National & International Case Team Delegate Jun. 2026 – Present
Wilfrid Laurier University *Waterloo, ON*

- Selected for both **Digital Strategy** and **LazSICC** case teams, representing Laurier both nationally and internationally; Competed in **15+** time-bound case competitions, earning **$3,000+** in competition winnings

Projects

Scholr | *Python, FastAPI, asyncio, OpenAI API, Pydantic, OpenAlex, MCP, Next.js, Postgres* tryscholr.com

- Built a recursive **multi-agent research system** over 200M+ papers: a planner decomposes queries into subtopics, parallel retrieval agents fan out via asyncio, a coverage-checker reflects on gaps and triggers re-retrieval; shipped to **300+ users**
- Engineered **self-correcting synthesis**: agents validate citations post-generation, a replanning loop injects failed-query history across 3 retries, and LLM-inferred intent detection routes follow-ups back into the agent loop with full session context
- Exposed via **MCP**, enabling other agents to invoke Scholr as a research tool; also ships as a full-stack web app with inline citation pills tracing every claim to its peer-reviewed source, BibTeX export, Google OAuth, and SSE token streaming

Studeal | *Python, FastAPI, LangGraph, Next.js, PostgreSQL + pgvector, Redis, Celery*

- Designed a multi-stage, **LangGraph-based agentic retrieval system** for deal discovery (retrieval, extraction, scoring), parallelizing search, scraping, and inference with semantic deduplication via pgvector
- Built a scoring agent with market context for daily personalized deal matching from natural-language watchlists; achieved **90%+ link-resolution accuracy** reconstructing missing product URLs via LLM-guided extraction
- Built the full stack with rate-limited APIs, Celery scheduling, Stripe access control, and a Next.js UI

PantryPal | *Python, FastAPI, Llama 3.2, PyTorch, QLoRA, DPO, scikit-learn, XGBoost* Blog Post

- Fine-tuned Llama 3.2 3B via **QLoRA SFT** on 8K constraint-labeled recipes; achieved **99.69% keyword avoidance** on a 500-prompt held-out benchmark vs. **54.92%** for base Llama 3.2 3B and **56.79%** for GPT-4o at only **$8 training cost**
- Studied whether domain-specialized weak models can replace human preference labelers to steer a stronger LLM; used **18 XGBoost classifiers** to generate **1,200+ DPO preference pairs** at **59% contrast rate** across 18 dietary categories

Technical Skills

Languages: Python, SQL (PostgreSQL, SQLite), TypeScript, JavaScript, HTML/CSS, C
AI / ML: PyTorch, LangChain, LangGraph, Hugging Face, MCP, fine-tuning, RAG, AI evals, OpenAI and Anthropic APIs
Frameworks: React, Next.js, FastAPI, Flask, Node.js
Libraries: pandas, NumPy, scikit-learn, Pydantic, SQLAlchemy, Selenium, BeautifulSoup, Matplotlib, Playwright
Tools: Git, Docker, Redis, Bash, Claude Code, GCP, Azure
Awards: President's Gold Scholarship, Loblaw National Scholarship, Stanford ML Specialization